Project Brief: Deep Reinforcement Learning for Automated Stock Trading

The Hook

Reinforcement learning (RL) represents the absolute bleeding edge of algorithmic trading, shifting the paradigm from static rule-based models to dynamic agents that learn by continuously interacting with the market. This paper outlines a powerful ensemble strategy that combines three state-of-the-art continuous action algorithms (PPO, A2C, and DDPG) to maximize portfolio returns while strictly penalizing risk. Mastering this architecture bridges the gap between theoretical artificial intelligence and production-grade automated execution systems used by top-tier quantitative funds.

Definition of Done (Project Scope)

For an advanced team, a fully completed project should include:

- **A cleanly structured codebase:** An object-oriented Python repository establishing a custom trading environment using the OpenAI Gym (now Gymnasium) framework.
- **The Trading Environment:** A defined state space (containing account balance, held shares, and technical indicators), an action space (buy, sell, hold continuous quantities), and a reward function driven by changes in portfolio value.
- **The RL Agents:** Implementation or integration of the three underlying actor-critic algorithms: Proximal Policy Optimization (PPO), Advantage Actor Critic (A2C), and Deep Deterministic Policy Gradient (DDPG).
- **The Ensemble Logic:** A rolling-window evaluation module that tests the three agents over a validation period and dynamically selects the one with the highest Sharpe ratio to trade the upcoming out-of-sample period.
- **Data Visualization:** A comprehensive cumulative return chart plotting the ensemble strategy against a baseline index (like the Dow Jones 30) and against the individual underlying models.
- **Performance Metrics:** Calculation of standard industry metrics, including annualized return, annualized volatility, maximum drawdown, and the Sharpe ratio.
- **Stretch Goal:** Take inspiration from Trutina and wrap the trained ensemble agent into a full-stack dashboard, allowing users to watch the agent visually execute trades and rebalance a portfolio over a historical holdout set.

9-Week Milestone Timeline

- **Week 1 (Feb 20 – Feb 26): Literature Review & RL Fundamentals**
  - Read the paper thoroughly. Ensure the team conceptually grasps Markov Decision Processes (MDPs) and how actor-critic architectures map states to actions while evaluating those decisions.

- **Week 2 (Feb 27 – Mar 5): Data Sourcing & Technical Indicators**
  - Set up the GitHub repository.
  - Download daily Open-High-Low-Close-Volume (OHLCV) data for the Dow Jones 30 constituents and compute the necessary technical indicators (MACD, RSI, CCI, ADX) used as state features.
- **Week 3 (Mar 6 – Mar 12): Building the Gym Environment**
  - Code the custom Gymnasium trading environment. This is the hardest foundational step; the agent needs rules for transaction costs, starting capital, and how actions directly impact the portfolio state.
- **Week 4 (Mar 13 – Mar 19): Integrating Baseline Agents**
  - Set up the PPO, A2C, and DDPG agents to interact with the custom environment. Ensure the action outputs correctly map to portfolio weight allocations.
- **Week 5 (Mar 20 – Mar 26): Training the Individual Models**
  - Run the training loops over the historical training data split. Monitor the episodic reward curves to ensure the agents are actually learning and converging rather than taking random actions.
- **Week 6 (Mar 27 – Apr 2): Developing the Ensemble Strategy**
  - Code the rolling-window logic that validates the models every few months (e.g., quarterly) and seamlessly passes trading control to the highest Sharpe ratio agent for the next quarter.
- **Week 7 (Apr 3 – Apr 9): Backtesting & Risk Management**
  - Run the full ensemble model over the out-of-sample test period.
  - Generate tearsheets to extract maximum drawdown and volatility metrics to ensure the agent isn't taking on catastrophic risk.
- **Week 8 (Apr 10 – Apr 16): Competition Readiness & Optimization**
  - Finalize the model inference pipeline so the agent can quickly generate portfolio allocations based on new daily data, ensuring the architecture is locked in for the Rutgers Quantitative Finance Club algorithmic trading competition on April 18th.
- **Week 9 (Apr 17 – Apr 23): Documentation & Portfolio Polish**
  - Finalize the GitHub README, clean up the codebase, and ensure the mathematical logic behind the reward function is heavily commented for recruiter review.

The Toolkit

- **Data:** The yfinance API to pull daily OHLCV data for the Dow Jones 30 components.
- **Libraries:**
  - gymnasium to construct the standard RL environment interface.
  - stable-baselines3 for highly optimized, out-of-the-box implementations of PPO, A2C, and DDPG (this prevents them from spending months debugging pure PyTorch tensor math).
  - stockstats or pandas-ta for easily calculating the technical indicators.

    - pyfolio or quantstats to instantly generate professional-grade backtesting tearsheets and risk metrics.